23MCT406
ENVIRONMENTAL SCIENCE

Module 3 - Water and Wastewater

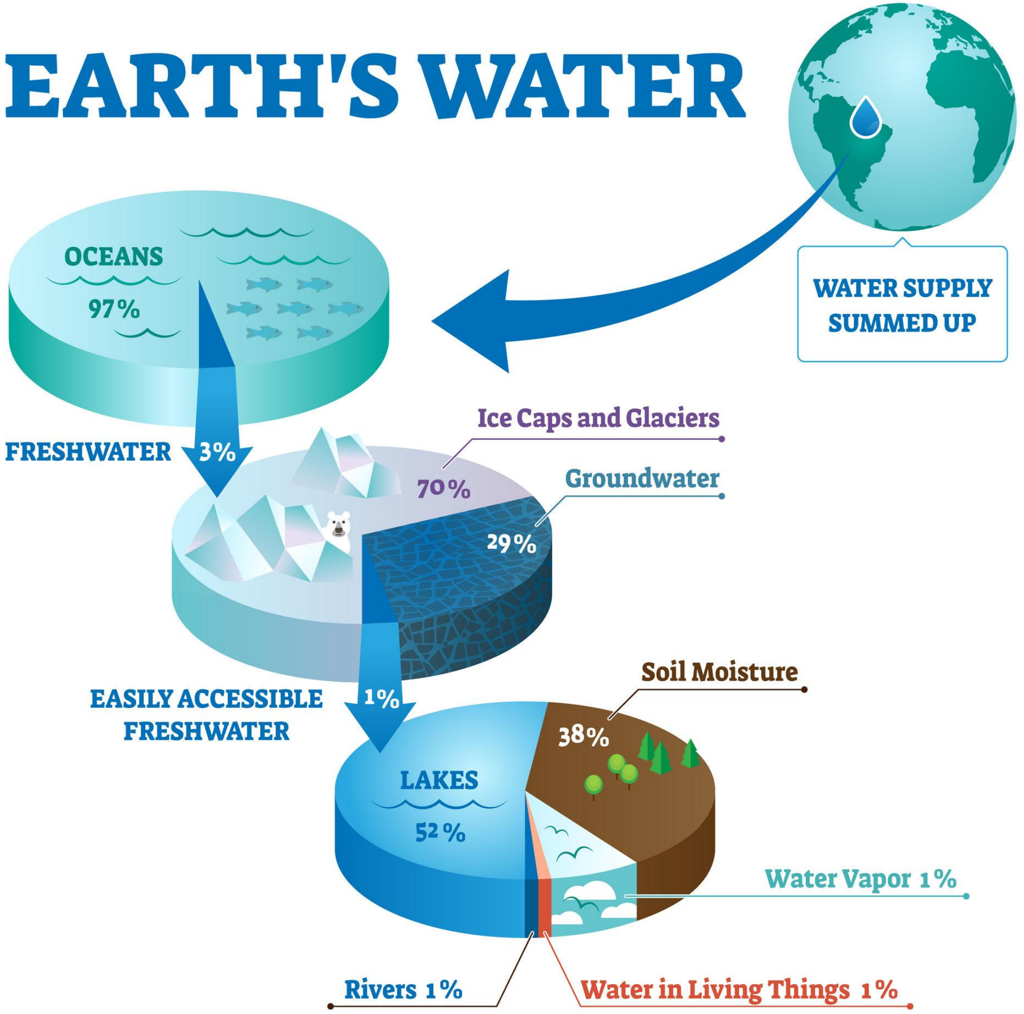
- Sources and availability of freshwater
- Water conservation strategies
- Water pollution and its impacts
- Water Quality Standards (IS 10500)
- Water quality index
- Overview of water treatment plant
- Sustainable water use and conflicts over water resources
- Wastewater sources and quality
- Wastewater disposal
- Oxygen sag curve
- Applicable wastewater discharge standards and typical flow schemes for sewage treatment plant
- Decentralized wastewater treatment- natural methods of wastewater treatment

WATER

- A single water molecule is composed of two atoms of hydrogen attached to one atom of oxygen (H_2O) by covalent bonds
- About 71 percent of the Earth's surface is water-covered
- **Oceans** hold about 96.5 percent of all Earth's water
- Water also exists in the air as **water vapor**, in **rivers** and **lakes**, in icecaps and **glaciers**, in the ground as soil moisture and in **aquifers**, and even in you and another organism

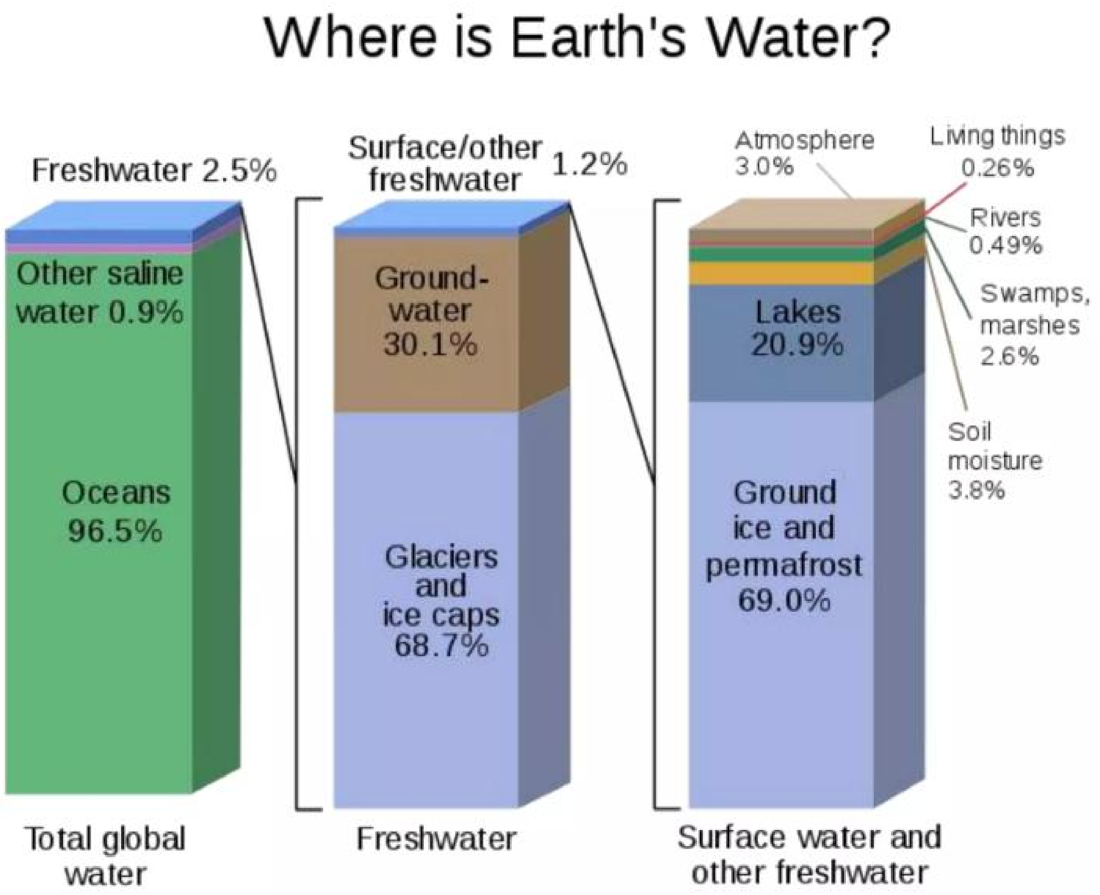
Sources and availability of freshwater

- Water resources are sources of water that are useful or potentially useful
- Uses of water include agricultural, industrial, household, recreational and environmental activities
- Virtually all of these human uses require fresh water



Contd…

- **97%** of the water on the Earth is **salt water**
- Only **3% is fresh water**
- slightly over two thirds of this is frozen in glaciers and polar ice caps
- The remaining unfrozen freshwater is found mainly as groundwater, with only a small fraction present above ground or in the air



SOURCES OF FRESH WATER

1. Surface Water
2. Under River Flow
3. Groundwater
4. Frozen Water
5. Desalination

1. Surface Water

- Surface water is water in a river, lake or fresh water wetland

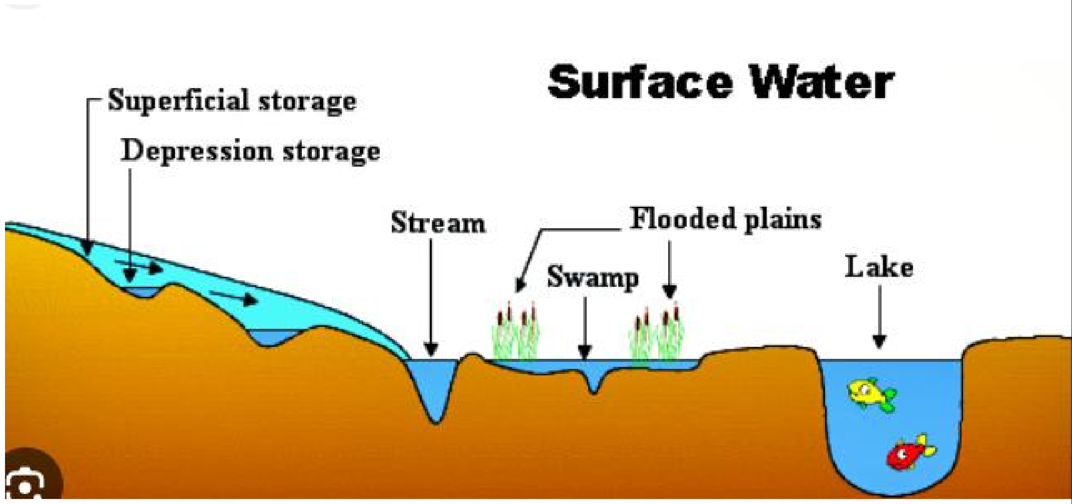
- Surface water is naturally replenished by:

➢ Precipitation and naturally lost through discharge to the oceans

➢ Evaporation

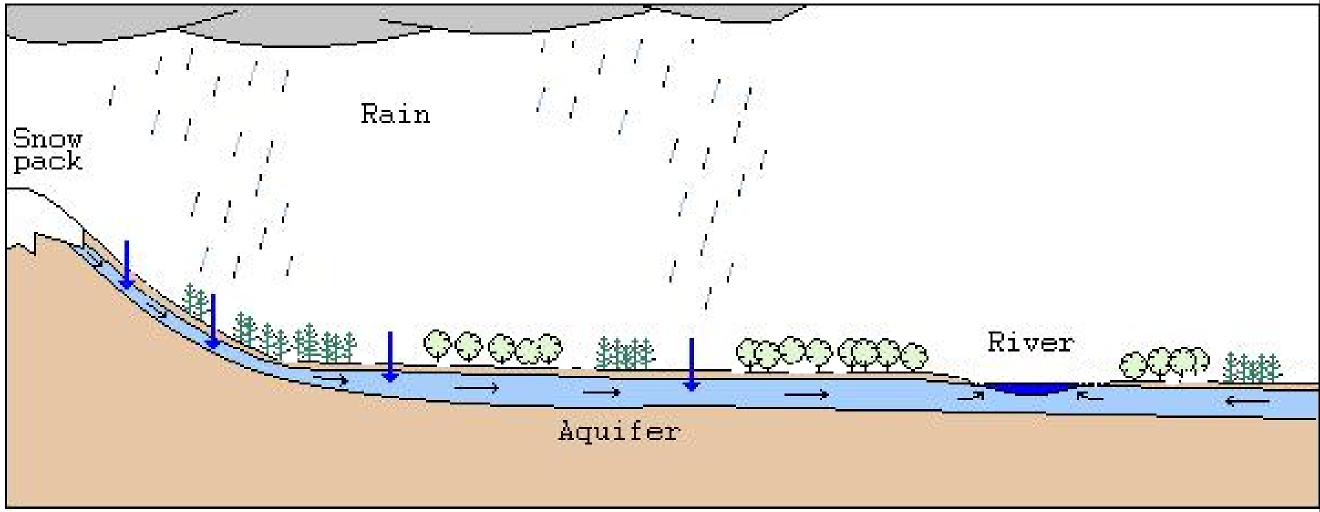
➢ Evapotranspiration

➢ Groundwater recharge



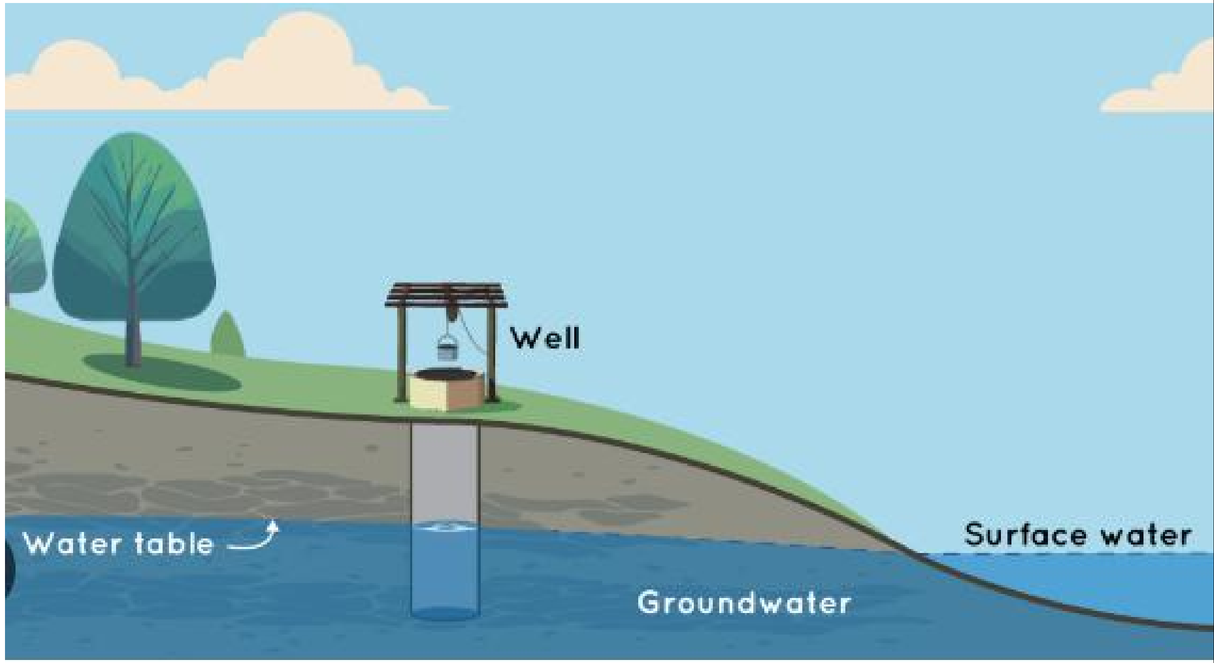
2. Under River Flow

- Under river flow is the water that flows through the rocks and sediments below a river. This flow is often called the **hyporheic zone**
- The hyporheic zone is the area below a river were water flows through **rocks and sediments**
- The hyporheic zone can be a significant part of a river's flow, especially in large valleys
- The hyporheic zone can exchange water with aquifers, which are large underground areas that store water



3. Groundwater

- Groundwater is fresh water located in the subsurface pore space of soil and rocks
- It is also water that is flowing within aquifers below the water table
- Sometimes it is useful to make a distinction between groundwater that is closely associated with surface water and deep groundwater in an aquifer (sometimes called "fossil water")

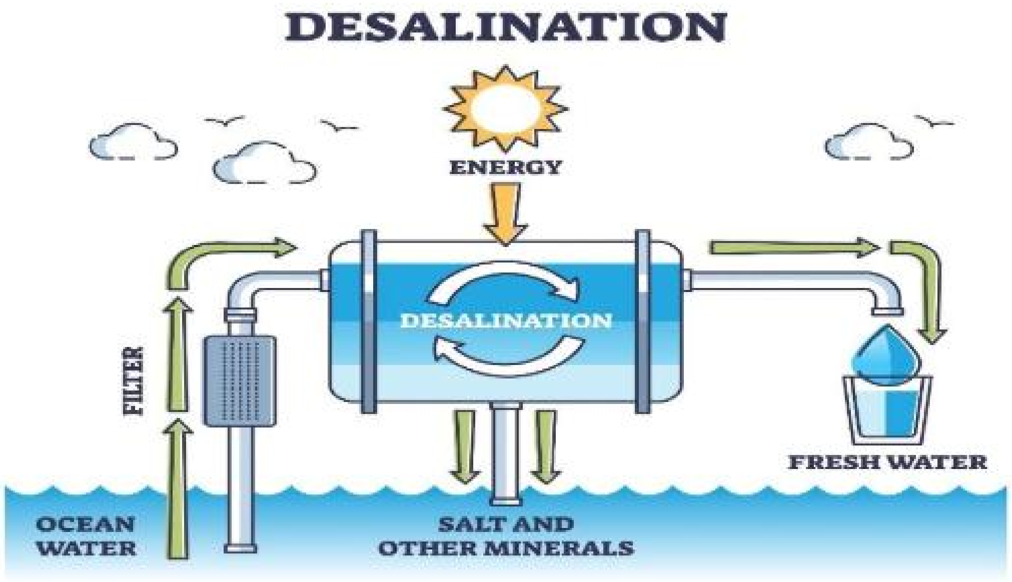


4. Frozen Water

- Frozen water, primarily in the form of glaciers and ice caps, holds about 68% of the Earth's fresh water
- As climate change accelerates ice melt, this frozen water becomes a significant source of fresh water for rivers and lakes
- In regions like the Himalayas and the Andes, glacial meltwater sustains agriculture and drinking supplies for millions
- However, excessive melting can lead to water scarcity in the long term as glaciers shrink

5. Desalination

- Desalination is an artificial process by which saline water (generally sea water) is converted to fresh water
- The most common desalination processes are distillation and reverse osmosis
- Desalination is currently expensive compared to most alternative sources of water, and only a very small fraction of total human use is satisfied by desalination



Water Conservation

- The process of saving water for future utilization is called water conservation
- With the increase in population, quality & quantity of water has declined

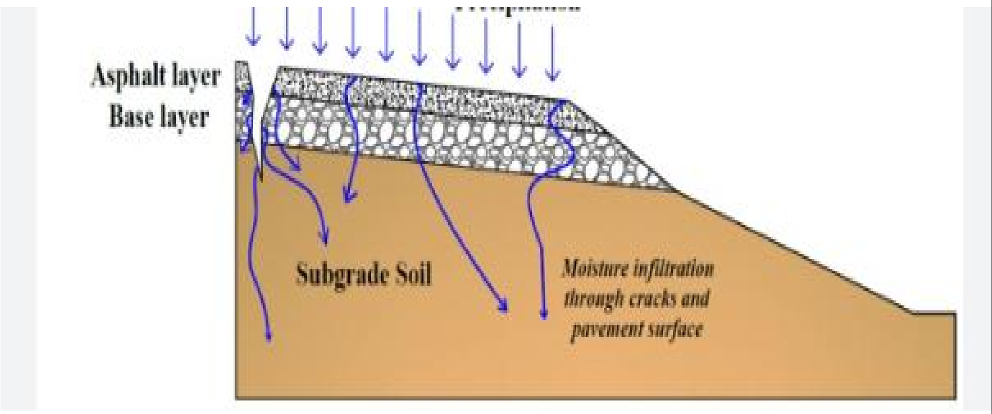
Water conservation strategies

1. Reducing evaporation
2. Reducing irrigation loss
3. Re-use of water
4. Preventing wastage of water
5. Decrease runoff loss
6. Avoid discharge of sewage

Contd…

1.Reducing evaporation loss

- Evaporation of water is commonly found in humid regions
- It can be reduced by placing horizontal barriers of asphalt below the soil surface, which increases water availability and crop yield



Contd…

2. Reducing irrigation loss

- Water loss during irrigation can be reduced by the following methods

a. Growing hybrid crop varieties

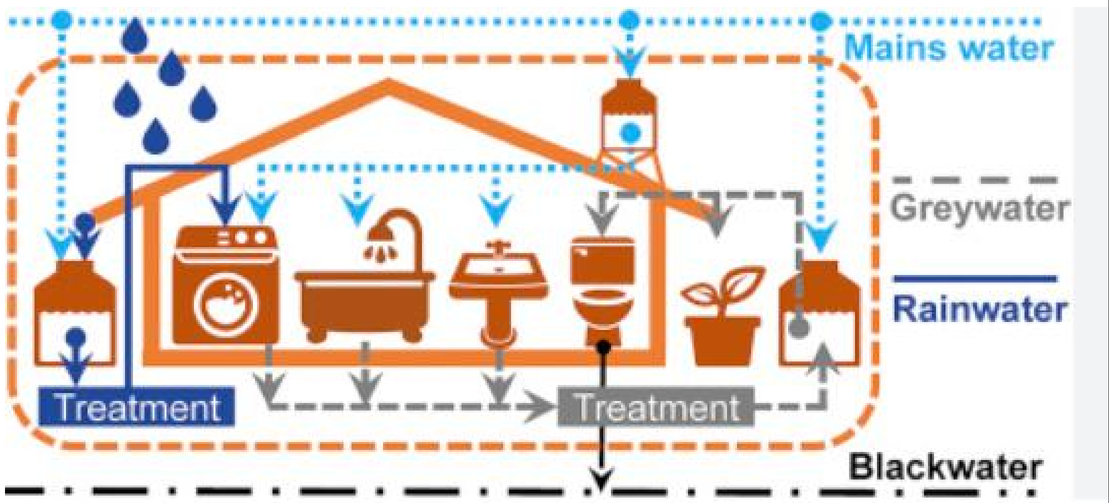
b. Irrigation in early morning or later evening

c. Using drip irrigation or sprinkler irrigation

Contd…

3. Re-use of water

- Treated water can be used for ferti-irrigation
- Grey water can be used for washing cars, watering gardens, etc
- Water used for washing vegetables and fruits can be used to water plants



Contd…

4. Preventing wastage of water

Wastage of water can be prevented by:

- Closing the taps when not in use
- Repair any leakage from pipes
- Use small capacity taps

Contd…

5. Decrease run-off loss

- Run-off can be reduced by allowing water to infiltrate into the soil
- This can be done by contour cultivation or terrace farming

Contd…

6. Avoid discharge of sewage

- Discharge of untreated sewage into natural water resources must be prevented

Methods of water conservation

- There are many methods available for conservation of water, some of the important measures are:

1. Rainwater harvesting
2. Watershed management

1.Rainwater harvesting

- Rainwater harvesting is a technique of collecting and storing rainwater in natural reservoirs or tanks for future utilization

Objecting of rainwater harvesting:

a. To meet increasing demands of water

b. To reduce ground water contamination

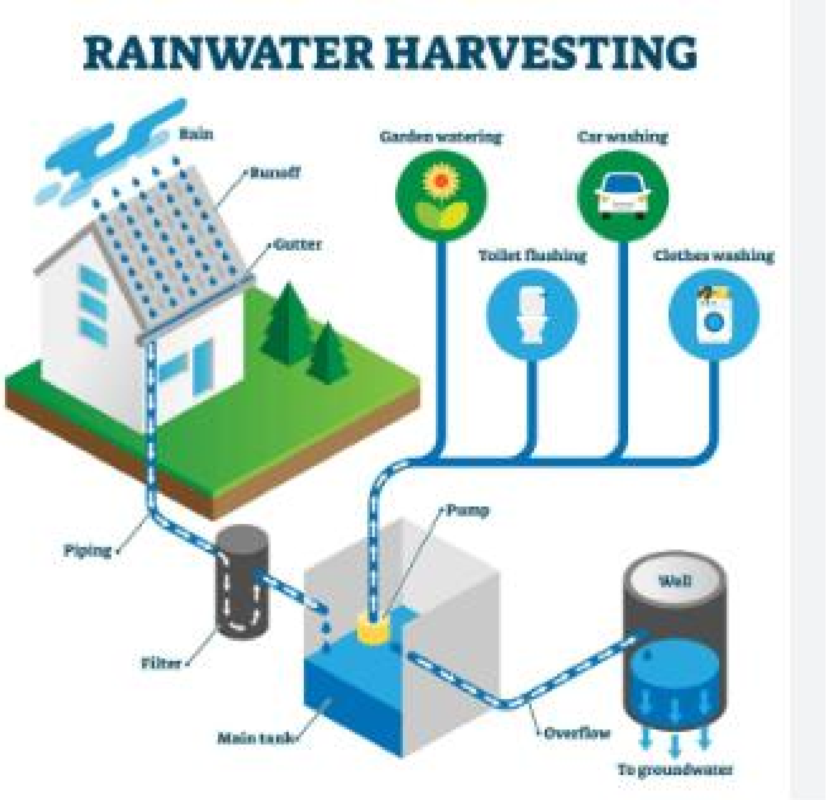
c. To minimize eater crisis

d. To reduce soil erosion

e. To raise the water table by recharging the ground water

ROOFTOP RAINWATER HARVESTING

- It is a method of collecting rainwater from the roof of the building and storing it for future use
- It is a low cost and effective technique for urban buildings



ADVANTAGES OF RAINWATER HARVESTING

- Improves the quality and quantity of ground water
- Reduces soil erosion and flooding
- Increases the availability of water
- Promotes water and energy conservation

2. Watershed management

Watershed

Land area from which water drains under the influence of gravity into a stream, lake, reservoir or other body of surface water

Watershed management

The management of rainfall and resultant runoff is called watershed management

Contd…

Factors affecting watershed:

- Overgrazing
- Deforestation
- Mining
- Construction activities
- Droughty climates

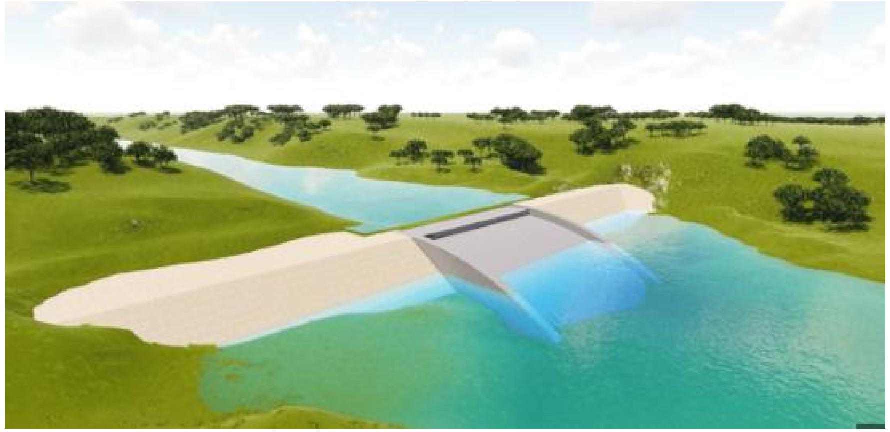
Objectives of watershed management

- To minimize the risks of floods, droughts and landslide
- To prevent soil erosion by runoff
- To raise the ground water level
- To generate employment opportunities
- To develop rural areas in the region

Watershed management techniques

1. Trenches: Trenches are dug at equal intervals to improve groundwater storage

2. Earthen dam: It is constructed in the catchment area to check the water run-off



Contd…

3. Farm pond: It can be built to improve water storage capacity of the catchment area

4. Dykes :It must be built along nullahs to raise the water table

WATER POLLUTION

- Water pollution can be defined as the presence of some foreign substances or impurities in water in such a quantity so as to constitute a health hazard, thus making water unfit for use

Contd….

- Water contained in water bodies like lakes, rivers and oceans are called surface water
- Water stored in aquifers (underground rock structures) is called underground water (subsurface water)
- Both surface and underground surface (subsurface water) are prone to pollution

The Water (Prevention and Control of pollution) Act, 1974

- The Water Act, 1974 is the first Indian law focused on the **prevention of discharging domestic and industrial wastewater into water bodies without treatment**
- This Act provides for the establishment of pollution control boards at Centre and States to act as watchdogs for prevention and control of pollution
- The Act was passed by the Parliament under Article 252 of constitution of India and was amended in 1978 and 1988

Contd…

Objectives of Water Act

- Prevention and control of water pollution
- Maintaining or restoring the wholesomeness of water
- Establishment of boards of the prevention and control of water pollution

Contd…

According to the Water Act, 'pollution' is defined as contamination of water or alteration of the physical, chemical or biological properties of water, or the discharge of any sewage or trade effluent (directly or indirectly) which is likely to render such water harmful or injurious to:

- Public health or safety
- Domestic, commercial, industrial, agricultural or other uses
- Life and health of plants, animals or aquatic organism

Sources of Water Pollution

1. Point Source

- Sources which can be readily identified at a single location(direct discharges from a single point)
- Some of the examples are wastewater discharge from industries, wastewater discharge from the domestic sector, etc.

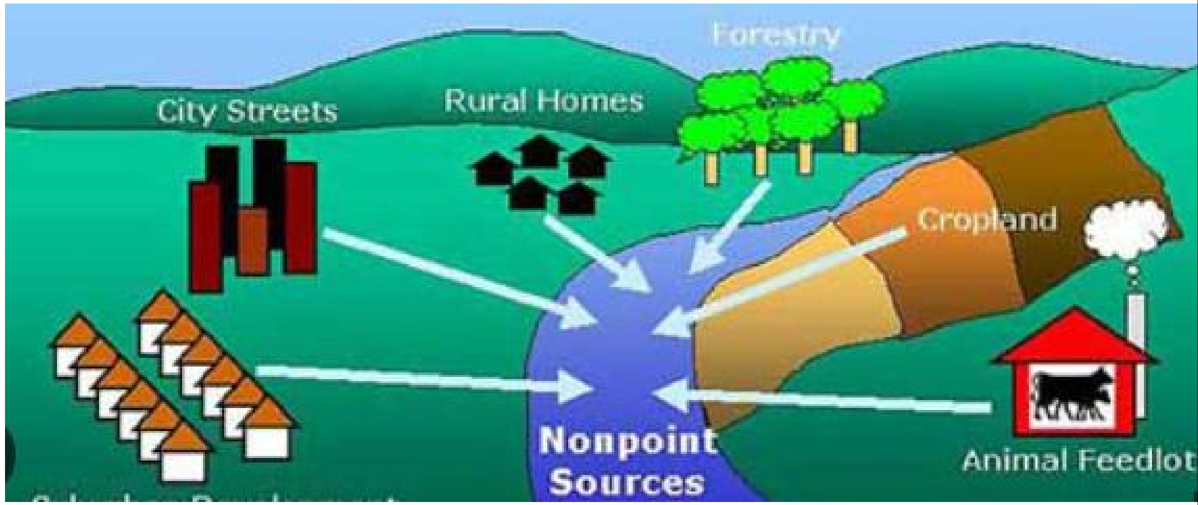
Contd…

- This type of discharges can be controlled easily
- Water pollution caused by these sources can be minimized if the effluents from these sources are controlled, treated up to acceptable levels and disposed off

Contd…

2. Non-point Source or Diffused Sources

- Non-point sources are diffused across a broad area
- Their source of origin cannot be traced to a single discharge point (whose location cannot be easily identified)
- Examples include run off from agriculture lands, mining area etc.
- It is very difficult to control pollution from non- point sources



Contd…

Natural Sources of Water Pollution

- Rain water
- Atmosphere
- Underground rocks
- Volcanoes

Anthropogenic Sources of Water Pollution

- Oil spills
- Industrial wastewater discharges
- Solid-waste disposal
- Runoff from agricultural fields
- Wastewater from automobile garages, etc

In India, water pollution comes mainly from the following three sources:

- Domestic wastewater
- Agricultural Runoff
- Industrial Effluent

1. Domestic Wastewater

- Domestic sewage is the wastewater generated from the household activities (residential community) that originates from residential areas, commercial places, institutions and other public places
- Generally domestic sewage consists of 99.9% of water and 0.1% of solids
- These solids are mostly organic (which is extremely putrescible) with a small fraction of inorganic matter

Contd…

- Its decomposition produces large quantities of malodorous gases and contains a number of pathogens (harmful micro-organisms)
- Domestic wastewater can be divided into two categories:

1. Black water (wastewater containing human excreta and urine)
2. Grey water (wastewater generated from bathroom and kitchen)

2. Agricultural Wastewater

- This is the run-off from the agricultural fields and animal farms
- This wastewater is considerably **rich in Nitrogen, Phosphate, Organic matter and Pesticides**
- Water bodies enriched with nutrients (Nitrogen and Phosphate) induce **rapid growth of microscopic plant life (algae)** in surface waters which lead to algal bloom thus **reducing the oxygen content** in the aquatic environment popularly known as Eutrophication (which is most common in stagnant water bodies such as ponds and lakes).

3. Industrial Wastewater

- Industrial wastewater (effluents) are the one which results from industrial operations
- Every year, the world generates around 400 billion tons of industrial waste, much of which is discharged into rivers, oceans and other waterways
- Industrial wastewater may have pollutants of almost all kinds ranging from simple nutrients and organic matter to complex toxic substances
- The nature and composition (characteristics) of the industrial wastewater vary widely from industry to industry, and even for the same industry depending upon: Raw materials, Processes & Operational factors

GROUNDWATER POLLUTION

- Water pollution is not confined to surface water alone
- Various kinds of harmful materials like fertilizers, pesticides, metals, etc present in solidwaste (disposed on land) get dissolved or leaked into water
- During rain these pollutants present in water percolate down into the soil and contaminates the groundwater beneath

WATER POLLUTANTS

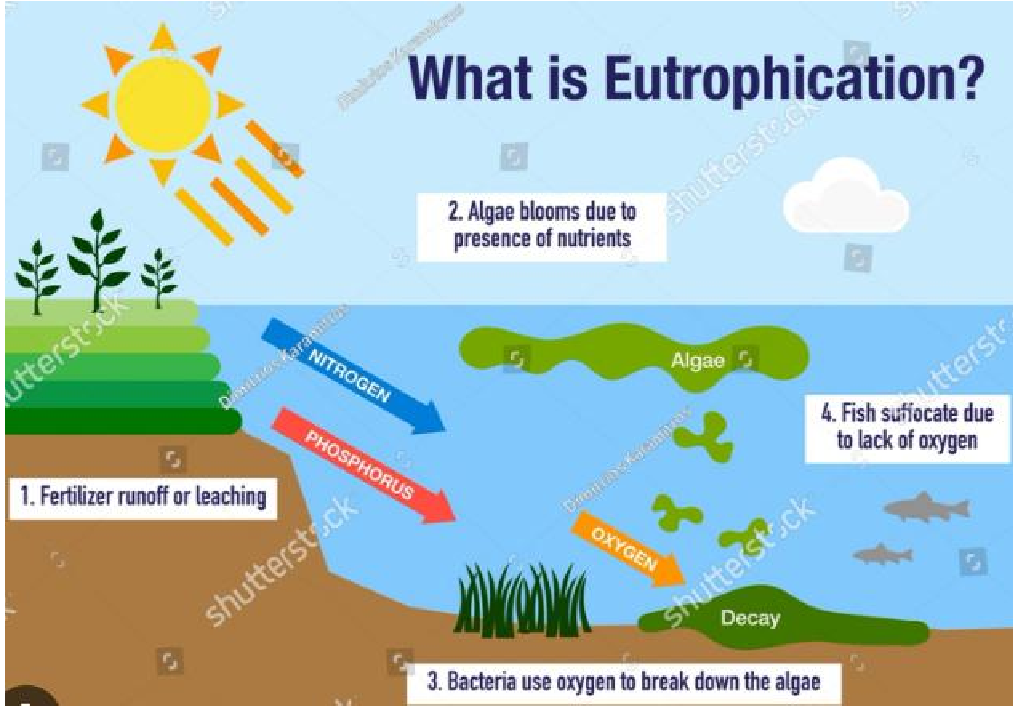
- Water pollutants are substances which causes water pollution

Contd…

Some of the commonly four pollutants in water are:

1) Inorganic salts
2) Acids/Alkalis
3) Organic matter
4) Suspended solids
5) Floating matter
6) Thermal discharges
7) Colouring materials
8) Toxic chemicals
9) Micro-organisms
10) Radioactive materials
11) Foam producing matter etc.

1.Inorganic salts



- Include nitrates, phosphates, carbonate chlorides, sulphates of Ca, Mg, Fe etc.
- These salts makes water "hard“
- Hard water increases soap consumption and deposits scales on pipelines
- Iron causes spots and stains on white fabrics
- Water bodies enriched with nitrates and phosphates induce rapid growth of microscopic plant life which can lead to algal bloom popularly known as Eutrophication

2. Acids/Alkalis

- Most freshwater bodies have a natural pH in the range of 6 to 8
- Acids and alkalies are discharged into streams by chemical industries and other industrial plants
- High concentrations of acid, sufficient to lower the pH to below 7.0 causes eye irritation to swimmers, rapid corrosion of ship's hull, and deterioration of fishermen's nets
- When the pH of water goes below 5, fish populations begin to disappear.
- Alkalies (e.g. NaOH) appear in wastewater from soap manufacturing, textile dyeing, rubber reclamation, leather tanning etc.
- High pH is also detrimental to aquatic life

3.Organic matter

- Organic pollution occurs when large quantities of organic compounds which act as substrates (food) for microorganisms, are released into water sources
- During the decomposition of organic pollutants, the dissolved oxygen in the receiving water may be used up at a greater rate than it can be replenished, **causing oxygen depletion**
- The **minimum amount of dissolved oxygen in water** for sustenance of **fish life is 3 to 4 mg/L**

4. Suspended solids

- Suspended solids either settle to the bottom of remains suspended in the water body
- These solids increase the turbidity of the water body
- Solids that gets deposited on the banks decompose and causes odour nuisance

5. Floating matter

- These include oils, greases and other materials which float on the surface, and not only make the river unsightly, but also obstruct passage of light through the water
- Presence of floating matter hinders the self-purification process of water bodies as well

6. Thermal discharges

- Increase in the normal temperatures of natural waters is caused by nuclear power plant/boiler/industrial discharges (where water is used as a cooling agent)
- Since warm water is lighter than cold, stratification occurs in the water body, and this causes most of the aquatic life to retreat to stream bottom
- Also, bacterial action increases which in turn results in accelerated depletion of the stream's dissolved oxygen content

7. Colouring materials

- Colour in water is an indicator of pollution (which often invites public attention)
- Colour interferes with the transmission of sunlight into the stream, thus reducing its natural disinfection action
- Colour in water is contributed by the effluents discharged from textile industries, paper mills, tanneries, slaughter houses etc.

8. Toxic chemicals

- Highly complex both organic and inorganic toxic compounds produced by various chemical industries have proved extremely toxic to aquatic life
- These include cyanides, sulphides, acetylene, alcohol, etc.
- Many of these have a cumulative effect on the aquatic environment
- For example, hexavalent Chromium found in wet cement is water soluble and can soak into skin and produce an allergic reaction

9. Micro-organisms

- Domestic sector, food processing industries and slaughter houses, usually discharge wastes containing micro-organisms
- Diarrhoea and Pneumonia are the biggest killer diseases in children aged under five years in India
- Oral Rehydration Salts (ORS) with zinc supplementation is the correct treatment for diarrhoea (as per 2013 WHO recommendation). Some of the water-borne diseases caused by pathogens are tabulated above:
- The above micro-organisms mainly bacteria are of two types:

a) Micro-organisms which assist in the degradation of the organic matter

b) Micro-organisms which are pathogenic in nature

10. Radioactive materials

- Radioactive waste typically comprises of a number of radio isotopes (which are unstable) emitting ionizing radiation which can be harmful to humans and the environment
- This radiation is not readily detectable by the methods usually employed to determine the presence of contaminants in the environment
- Exposure to high levels of radioactive waste may cause mutation and cancer

11. Foam producing matter

Presence of foam producing matter leads to an undesirable appearance of the receiving streams / water bodies which is usually discharged by textile mills, pulp and paper mills and some chemical manufacturing industries.

- Some of the noticeable signs of water pollution are:
- Offensive odours from rivers, streams, lakes and oceans
- Oily and greasy material floating on the surfaces of water bodies
- Unchecked growth of aquatic weeds in water bodies
- Bad taste of drinking water
- Decrease of aquatic life in fresh water bodies

Impacts of water pollution

Water pollution can cause a variety of problems to human beings and other living things, some of them are listed below:

1. Consuming contaminated water leads to a number of health problems due to water-borne diseases (cholera, typhoid, jaundice, etc).

- Excess amount of metals in drinking water (fluoride, lead, copper, etc) can cause general body weakness, mental retardation, failure of internal organs, cancer and even death

2. Pollutants in water leads to widespread destruction of aquatic life

3. Nutrient rich (nitrogen and phosphate) wastewater leads to abundant growth of algae which results in eutrophication

4. Toxic pollutant in water, easily enters the food chain which results in bio-magnification [Bio-magnification is the increase in the concentration of a pollutant from one link to another in a food chain].

CONTROL MEASURES OF WATER POLLUTION

1. Drinking water should be boiled, cooled and then used
2. Disinfection of drinking water should be done by using chemicals like bleaching powder
3. Pesticides and insecticides should be prevented from nearby use of water lakes, ponds and pools
4. Drainage water should not be allowed to mix with drinking water
5. Drainage system should be maintained properly
6. Chlorination process is to be adopted for drinking water.

Contd....

7. Appropriate wastewater treatment is to be provided before discharging wastewater from industries and domestic sectors
8. Using phosphate free detergents and reducing the use of fertilizers (in agriculture), the phenomenon of eutrophication can be controlled
9. The public should be made aware of the ill-effects of disposing wastewater onto waterbodies
10. Legislative measures have to be made more stringent and the principle of PPP (Polluter Pays Principle) should be strictly followed [i.e. polluter should bear the cost of cleanup operations, since polluter is responsible for pollution]

Water Quality Standards (IS 10500)

Characteristics	Parameters	Acceptable Limit	Permissible Limit
1. Physical	Temperature		
	Colour	5 Hazen	15 Hazen
	Taste	Agreeable	Agreeable
	Odour	Agreeable	Agreeable
	Total Dissolved Solids	500 mg/L	2000 mg/L
2. Chemical	pH	6.5-8.5	No relaxation
	Acidity		
	Alkalinity	200 mg/L	600 mg/L
	Hardness	200 mg/L	600 mg/L
	Chloride (Cl)	250 mg/L	1000 mg/L
	Nitrate (NO_3)	45 mg/L	No relaxation
	Iron (Fe)	0.3 mg/L	No relaxation
	Chromium (Cr)	0.05 mg/L	No relaxation

Characteristics	Parameters	Requirements
3. Biological	E. Coli or thermotolerant coliform bacteria	Shall not be detectable in any 100 ml sample

Water Quality Index (WQI)

- WQI indicates the quality of water in terms of index number
- It represents overall quality of water for any intended use
- It is defined as a rating reflecting the composite influence of different water quality parameters were taken into consideration for the calculation of water Quality index (WQI)

Contd…

- The indices are among the most effective ways to communicate the information on water quality trends to the general public or to the policy makers and in water quality management
- In formulation of water quality index the relative importance of various parameters depends on intended use of water
- Mostly it is done from the point of view of its suitability for human consumption